



Stochastic Modeling of Glacial-Induced Water Level Rise via Monte Carlo Simulations.

Course: CSE400 – Fundamentals of Probability in Computing
Domain: Climate

GROUP-12

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Project Concept & Motivation

- Glaciers are very sensitive indicators of climate change and a key contributor to global sea level rise.
- Glacier melting is dependent on several climatic variables which are uncertain.
- Deterministic models give single estimates and miss extreme but likely outcomes.
- Probabilistic modeling allows uncertainty and risk to be quantified.
- Monte Carlo simulations make it possible to estimate Probability Density Functions (PDF's) rather than point predictions.



Background

- The Surface Energy Balance (SEB) framework is generally used to model glacier mass balance.
- The melt behaviour is determined by many factors, including Precipitation, Air temperature, Ice Albedo, Snowfall Threshold and Cloudiness.
- Each SEB model depends on several input processes, each with its own uncertainty.
- Our research paper identifies melting of glaciers and ice sheets as the greatest source of uncertainty in predicting sea-level rise in the future.



Mathematical Basis

- Glacier mass balance MM is governed by Precipitation, Air temperature, Ice Albedo, Snowfall Threshold and Cloudiness.
 - Small uncertainties in inputs can cause large variations in outputs.
 - This sensitivity motivates a stochastic rather than deterministic approach.
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Problem Statement

To predict global sea-level rise by calculating how much glaciers melt using a randomized Monte Carlo simulation. Instead of one 'best guess,' we will run 5,000 simulations where we randomly vary five key factors—Air Temperature, Precipitation, Ice Albedo, Snowfall Threshold, and Cloudiness—to see which one causes the most melting. Our goal is to create a probability map (PDF) that shows the risk of extreme sea-level rise under different global warming scenarios.

KEY GOALS:-

- Model uncertainty in real time inputs.
- Identify which parameters contribute most to output variance.
- Estimate PDFs and CDFs of water level rise.

Project System Overview

Input:

- Climate and glaciological variables (Precipitation, Air temperature, Ice Albedo, Snowfall Threshold and Cloudiness.)

Processing (Monte Carlo Module):

- Random sampling of input variables and simulation of glacier melt using the energy balance model.

Model Core (Melt Estimation):

- Computation of glacier melt and runoff for each simulation run.

Output:

- Probability Distribution (PDF/CDF) of Water Level Rise and Risk Score.

Key Random Variables

- Air Temperature - The primary melt driver
- Precipitation - Directly affects mass balance
- Ice Albedo - Surface reflectivity
- Snowfall Threshold - Determines if precipitation is rain or snow
- Cloudiness - Blocks solar radiation while trapping heat

Modeling Air Temperature Uncertainty

$$T_{\text{sim}}(t) = T_{\text{meas}}(t) + \epsilon_s + \epsilon_r(t)$$

Meaning of Terms

- **$T_{\text{sim}}(t)$** : Simulated air temperature used in the melt model
- **$T_{\text{meas}}(t)$** : Observed / baseline temperature from data
- **ϵ_s** (Systematic error):
 - Constant bias for one Monte Carlo run
 - $\epsilon_s \sim N(0, 0.3^\circ\text{C})$
- **$\epsilon_r(t)$** (Random error):
 - Hourly atmospheric variability
 - $\epsilon_r(t) \sim N(0, 0.3^\circ\text{C})$

Meaning of Terms

- Real temperature contains long-term bias and short-term fluctuations
- Systematic error models sensor/calibration uncertainty
- Random error models natural weather noise
- This separation ensures realistic uncertainty propagation in glacier melt
- Temperature uncertainty directly affects melt rate and water-level rise

Probabilistic Assumptions

1. Normal Distribution of Errors

We assume that uncertainties follow a normal (Gaussian) distribution with mean zero. This means the true value is usually close to the measured value but can vary slightly in both directions.

2. Independence of Variables

We assume that the uncertainties in temperature, precipitation, albedo, cloudiness, and other inputs are independent and do not directly affect each other.

3. Dominance of Systematic Errors

We assume that long-term systematic errors, such as sensor bias or calibration issues, have a greater impact on glacier melt than short-term random noise.

4. Non-Linear Sensitivity

Our model assumes that uncertainty does not increase linearly. Small changes in inputs, such as temperature, can lead to large changes in glacier melt due to non-linear physical processes.

5. Probabilistic Snow Threshold

Instead of using a fixed temperature to separate rain and snow, we assume a gradual transition between 1°C and 2°C, where both states are possible with certain probabilities.

6. Temporal Correlation

For variables like cloudiness, we assume that errors persist over time. This reflects real weather patterns, which change gradually rather than randomly every hour.



Thank You

